Al Balqa' Applied University
Faculty of Engineering Technology

كلية الهندسة التكنولوجيه

Student Name: .....

Mid Exam

Course Name: signal and system	Date: 12/4/2022
lecturer: Dr. Aws Al-Qaisi	Exam Time: 40 min

1-given x(t)=-3t+15 where, $4 \le t \le 5$. find the time interval for x(-2t+4), Select one:

a. $-0.8 \le t \le -0.4$ b. $-1.75 \le t \le -1.5$ c. $-0.75 \le t \le -0.5$ d. $0.5 \le t \le 0.75$ e. None

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2-given x(t)=t+4 where $-4 \le t \le -2$, find x(-2t+4)

a. x(t)= -4t+4 where, $1 \le t \le 1.5$ b. x(t)= -4t+6 where, $1 \le t \le 1.5$

c. x(t)= -4t+4 where, $1.5 \le t \le 2.5$ d. x(t)= -4t+6 where, $1.5 \le t \le 2.5$ e. None

3-if x(t)=r(t+3)-r(2t), the x(t) during the time intervals $-3 \le t \le 0$ can be written as

a. 3 b. t+3 c. 3-t d. t-3 e. None

4-x(t)=r(t)-2r(t-2) find x(-2t+2).

a. x(-2t+2)= 2 -2t (-1 <t <0), x(-2t+2)= 2t+2 (0<t <1)

b. x(-2t+2)= 2-2t (0< t <1) , x(-2t+2)= 2t+2 (-1 <t <0)

c. x(-2t+2)= 2t+2 (0 <t <1) , x(-2t+2)= 2t+2 (-2<t < 0)

d. x(-2t+2)=-2t+2 (0< t <1), x(-2t+2)=2t+2 (-1 <t < 0)

e. None

5-If x(n)=(0,0,1,2,3,4,0,0) then x(n-3) is

a. (0,0,0,2,4,6,8,0,0) b. (0,0,0,1,2,3,4,0,0) c. (1,2,3,4,0,0,0,0,0) d. (0.0,0,0,0,1,2,3,4)

6-The impulse response h(n) of a linear time invariant system is given by:

$h(n) = u(n+3)+u(n-2)+2u(n-7)$, where u(n) is the unit step sequence.

The above system is:?

a. stable and causal b. stable but not causal c. causal but unstable

d. unstable and not causal e. None

7-Determine the product of two signals: x1 (n) = {2,1,1.5,3}; x2 (n) = { 1,1.5,0,2}.

a. {2,1.5,0,6} b. {2,1.5,6,0} c. {2,0,1.5,6} d. {2,1.5,0,3}

8- h1(t)convolved with h2(t) = h2(t) convolved with h1(t)

a. Associative property b. Distributive property

c. Commutative property d. Multiplication property

9- find y(n)=x(n)*h(n), where x(n)=[1,1] and h(n)=[1,1]

a. [1,2,1] b. [1,2,3] c. [3,1,1] d. [1,1,2]

10- Find the fundamental period and frequency of the following signal

$x(t)=2\cos(6\pi t+ \pi/6)$

a. T=0.16 sec ,f=6 Hz **b.** T=10 sec ,f=0.1 Hz **c.** T=0.33 sec ,f=3 Hz **d.** T=5 sec ,f=0.2 Hz

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11- Is the following signal an energy signal

$x(t) = u(t)-u(t-1)+r(t-4)$

a. YES b. NO

12- Find the value of $2h[n]*\delta(n+1)$

a. 2h[n-2]. b. 2h[n]. c. 2h[n-1]. d. 2h[n+1].

13- $h1(t)*h2(t)*h3(t)= h3(t)*h2(t)*h1(t)$

this property of convolution process is called _____

a. Associative property b. Commutative property

c. Distributive property d. convolved with impulse property.

14- Noise generated by an amplifier of radio is an example for?

a. Discrete signal b. Deterministic signal c. Random signal d. Periodic signal

15- any time delay or time in advance in the input signal leads to a different time delay or time in advance in the output signal :

in this case the system is called

a. Time invariant systems b. Memory less systems

c. Causal systems d. message frequency

Question	1.	2.	3.	4.	5.	6.	7.	8.	9.	10
Answer										
Question	11	12	13	14	15					
Answer										

a. Time invariant systems

b. Memory less systems

c. Causal systems

d. message frequency

Question	1.	2.	3.	4.	5.	6.	7.	8.	9.	10
Answer	A	D	C	C	D	B	A	C	A	C

Question	11	12	13	14	15
Answer	A	D	B	C	A